

COF-System AI4S

Knowledge-driven intelligent assistant for COF design

This reviewer brief condenses the system demonstration video into a compact PDF. It shows how a user moves from COF questions and monomer choices to graph-traced answers, literature evidence, design analysis, and 3D molecular preview.


3

functional modules


508

monomer records


507

3D-ready records


771

COF/dimer candidates

PURPOSE

When the original video cannot be uploaded, the reviewer can still understand what the system does, what evidence is exposed, and how the interface supports COF design.



Knowledge QA

Natural-language COF question answering with graph triples, raw subgraph JSON, generated Cypher, and literature fragments.



COF Design Analysis

Node-linker or monomer-pair analysis with staged checks for reactivity, formability, stability, properties, and HER potential.



Monomer Library

Searchable monomer records with molecular metadata, 3D structure preview, and related synthesizable COF/dimer candidates.

REVIEWER READING PATH

Page 2 explains the dual-evidence architecture. Pages 3-5 show the three major interfaces. Page 6 turns the 516-second video into a concise storyboard.

Evidence Architecture

The system is organized around dual evidence: a literature-RAG path and a knowledge-graph retrieval path.



DUAL-EVIDENCE TRACEABILITY

Generated answers are not shown as black-box text. The interface places the final response next to the evidence used to produce it, so reviewers can inspect both the paper-level context and the structured domain relations.

LITERATURE RAG

Paper snippets ground the response.

KNOWLEDGE GRAPH

Triples show source, relation, and target.

VISIBLE AUDIT TRAIL

Inspect raw JSON and generated Cypher.

KNOWLEDGE QA WORKFLOW

Classify the question, parse entities, retrieve graph and literature evidence, then synthesize an answer.

COF DESIGN WORKFLOW

Resolve monomers, evaluate chemistry and properties, then return an evidence-supported design recommendation.

IMPLEMENTATION SIGNALS VISIBLE IN THE DEMO

The web interface includes dedicated pages for QA, COF design analysis, monomer library exploration, and settings. Neo4j provides structured graph relations; the literature retriever provides local paper fragments; the 3D viewer supports monomer and dimer visualization where structures are available.

1. Knowledge QA

A reviewer can ask a COF-related question in natural language and inspect how the answer is grounded in graph and literature evidence.

COF-System AI4S
v1.0.0

知识问答

COF 设计分析

单体库

系统设置

知识问答

自然语言问题、图检索、知识片段、可审计分析步骤

自然语言输入

Schiff-base 和 beta-ketoenamine 两种方式对 COF 的光催化析氢表现有什么影响?

发送并检索

分析步骤

6/6 done

问题归类

done

实体解析

done

知识库检索

done

图检索

done

证据组织

done

最终结论

done

最终答案

Schiff-base 与 β -ketoenamine 两种连接方式对 COF 的光催化析氢性能有显著影响。Schiff-base 反应形成的可逆共价键为构建有序框架提供了动态调控窗口,而不可逆的 β -ketoenamine 结构则赋予材料更稳定的化学稳定性。研究表明,通过羧基功能化修饰 β -ketoenamine-linked COF,可有效抑制光降解并延长载流子寿命,使析氢量子产率提升至 37% (未修饰体系仅 7.8%)。此外,调节 β -ketoenamine 与羧基的比例能优化框架有序度,增强可见光吸收能力,但过度增加刚性结构可能导致比表面积下降,反而削弱传质效率。目前证据表明,两种连接方式的协同作用可通过分子工程手段走向调控,但具体构效关系仍需更多实验验证。

图检索

关系	用实例
Electrodes	INVOLVE COF
environmental issues	INVOLVE COF
NF-kappa B activation	INVOLVE COF
copper coordination environment c...	INVOLVE COF
cleavage of H-Cl covalent bond	INVOLVE COF
coordination bond cleavage (Cu-N)	INVOLVE COF
structural microenvironment	INVOLVE COF
synthetic_methods	INVOLVE COF
mandible	INVOLVE COF
maxilla	INVOLVE COF
aqueous media	REACTION_CONDITIONS COF
phenol degradation	PHOTOCATALYTIC_ACTIVITY COF
catalyst	REUSABLE COF
for catalysis	REUSABLE COF
multiple reaction cycles	REUSABLE COF
PC-TpPa-1	MODIFIED_TO COF
MCOF	MODIFIED_TO COF
COF-CN	MODIFIED_TO COF

原始子图 JSON (已展开)

```
{
  "source": "nextj",
  "nodes": [
    {
      "id": "schiff Electrodes",
      "label": "Electrodes",
      "label": [
        "Result"
      ]
    },
    {
      "properties": {
        "value": "Electrodes"
      }
    }
  ],
  [

```

Cypher 查询 (已展开)

```
MATCH (a:Node)-[r]-(b:Node)
WHERE toLower(a.name) CONTAINS toLower("schiff") OR toLower(a.name) CONTAINS toLower("base") OR
toLower(a.name) CONTAINS toLower("schiff") OR toLower(a.name) CONTAINS toLower("base")
OR toLower(a.name) CONTAINS toLower("beta-ketoenamine") OR toLower(a.name) CONTAINS toLower("beta-
ketoenamine")
OR toLower(a.name) CONTAINS toLower("cof") OR toLower(a.name) CONTAINS toLower("cof")
toLower(a.name) CONTAINS toLower("photocatalytic") OR toLower(a.name) CONTAINS
toLower("photocatalytic")
OR toLower(a.name) CONTAINS toLower("hydrogen") OR toLower(a.name) CONTAINS toLower("hydrogen")
RETURN a.name AS source, type(r) AS relation, b.name AS target
LIMIT 100
```

知识库片段

1 / 5

KB-1

[KB-1] Activation of Carbonyl Oxygen Sites in β -Ketoenamine-Linked Covalent Organic Frameworks via Cyano Conjugation for Efficient Photocatalytic Hydrogen Evolution

1. Introduction · DOI: 10.1002/ami.202101017

Among various COFs, β -ketoenamine-linked COF, a class of $S-C(-N-)$ covalent frameworks, was first reported by Banerjee et al.[16] The preparation processes proceed with reversible Schiff-base reaction to construct an enol form with crystalline arrays and then irreversible enol-keto tautomerism to get the final keto-form networks, which exhibit the extraordinary chemical stability in boiling water, as well as in acidic H_2O and basic solutions $NaOH$ and basic solutions $NaOH$. Benefiting from these features, the ketoenamine-based materials have quickly evoked soaring interest in photocatalysis research field. The previous works were mainly committed to the morphology control, chemical stability, and charge transport of the COFs[17] Though the photocatalytic performance of β -ketoenamine-linked COFs has been improved to some e...

INPUT

Submit a natural-language COF question.

ANSWER

Read the synthesized response after retrieval.

TRACEABILITY

Inspect graph triples, Cypher, raw JSON, and paper fragments.

USE CASE

Use this page when the reviewer needs to see how COF knowledge questions are answered and traced.

COF-System AI4S reviewer demo PDF

3

2. COF Design Analysis

The design page evaluates whether a monomer combination is chemically plausible for COF construction and surfaces the evidence chain behind the recommendation.

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v1.0.0

知识问答

COF 设计分析

单体库

系统设置

COF 设计分析

单体组合、反应条件、COF 可形成性、稳定性、HER 潜力和证据链分析

COF 设计输入

节点选择: 单体库 人工输入 选择单体库 预览单体 3D 已显示

当前选中: Tp - 2,4,6-trihydroxybenzene-1,3,5-tricarbaldehyde - aldehyde

连接体: 单体库 人工输入 选择单体库 预览单体 3D 已显示

人工输入已解析: pPD, Pa-1, Pa - benzene-1,4-diamine - amine_primary: aniline

输入格式: auto 连接键: imine

溶剂: mesitylene/dioxane 1:1 反应类型: Schiff base 温度: 120 °C 时间: 72 h 氛围: N2

分析并检索 预览二聚体 3D 已显示

3D 二聚体 3D 预览

节点 3D 当前: Tp

连接体 3D 当前: pPD, Pa-1, Pa

COF 二聚体 3D Tp + pPD, Pa-1, Pa - imine

分析步骤 · CoT 领域分析链

8/8 done

1 单体分析 done

2 反应活性分析 done

3 COF 可形成性评估 done

4 稳定性分析 done

5 物理属性推断 done

6 HER 潜力评估 done

7 证据链组织 done

8 最终结论 done

最终答案

该组合具备构筑COF的化学基础。2,4,6-三羟基苯-1,3,5-三甲醛 (Tp) 与对苯二胺 (Pa-1) 通过醛胺缩合形成亚胺键, 符合Schiff base反应路径。两者连接数分别为3和2, 可构建二维层状或三维网络结构。知识库中已存在20个匹配COF记录, 包括TP-COF等典型体系, 证实其可形成性。亚胺键赋予材料较高结晶度, 理论比表面积可达数千m²/g量级。带隙可通过共轭程度调控, 尽管羟基可能引入局部极性, 但整体骨架稳定性良好, 适用于光催化产氢。若需提升HER性能, 建议结合助催化剂或掺杂策略优化微环境。实验上推荐采用均相溶液法, 以mesitylene/dioxane混合溶剂促进有序组装, 并通过PXRD验证长程有序性。

图谱证据

头实体	关系	尾实体
2,4,6-trihydroxybenzene-1,3,5-tricar...	IDENTIFIES_MONOMER	Tp
benzene-1,4-diamine	IDENTIFIES_MONOMER	pPD, Pa-1, Pa
TP-COF	SYNTHESIZED_FROM	Tp
TP-COF	SYNTHESIZED_FROM	pPD, Pa-1, Pa
Tp	SYNTHESIZES	TP-COF
pPD, Pa-1, Pa	SYNTHESIZES	TP-COF
TP-COF	SYNTHESIZED_FROM	Tp
TP-COF	SYNTHESIZED_FROM	pPD, Pa-1, Pa
Tp	SYNTHESIZES	TP-COF
pPD, Pa-1, Pa	SYNTHESIZES	TP-COF
DP-COF-1	SYNTHESIZED_FROM	Tp
DP-COF-1	SYNTHESIZED_FROM	pPD, Pa-1, Pa
Tp	SYNTHESIZES	DP-COF-1
pPD, Pa-1, Pa	SYNTHESIZES	DP-COF-1
TIQ2 - TpPa - 1 - COF	SYNTHESIZED_FROM	Tp
TIQ2 - TpPa - 1 - COF	SYNTHESIZED_FROM	pPD, Pa-1, Pa
Tp	SYNTHESIZES	TIQ2 - TpPa - 1 - COF
pPD, Pa-1, Pa	SYNTHESIZES	TIQ2 - TpPa - 1 - COF

原始子图 JSON (已展开)

```
{
  "source": "neo4j-cofdesign",
  "nodes": [
    {
      "id": "168891450-650-410-617-4507a52a51a593",
      "label": "2,4,6-trihydroxybenzene-1,3,5-tricarbaldehyde",
      "labels": [
        "TPAC",
        "BNode"
      ],
      "properties": {
        "node_type": "TPAC",
        "monomer_id": "B000370",
        "label": "2,4,6-trihydroxybenzene-1,3,5-tricarbaldehyde"
      }
    }
  ]
}
```

Cypher 查询 (已展开)

```
MATCH (t1:TPAC {value:'2,4,6-trihydroxybenzene-1,3,5-tricarbaldehyde'})->(t2:IDENTIFIES_MONOMER)->(a1:Monomer)
MATCH (t2:TPAC {value:'benzene-1,4-diamine'})->(t3:IDENTIFIES_MONOMER)->(a2:Monomer)
MATCH (cof:COF)->(s1:SYNTHESIZED_FROM)->(a1)
MATCH (cof)->(s2:SYNTHESIZED_FROM)->(a2)
OPTIONAL MATCH (a1)->(s3:SYNTHESIZED_FROM)->(cof)
OPTIONAL MATCH (a2)->(s4:SYNTHESIZED_FROM)->(cof)
RETURN a1, s1, a1, i2, s2, a2, cof, s3, s4, s5, s6
LIMIT 30
```

知识库片段

1 / 5

KB-1 Regioisomeric Benzotriazole-Based Covalent Organic Frameworks for High Photocatalytic Activity

CONCLUSIONS · DOI: 10.1021/acs.catal.3c04145

Preparation of Monomers and COF Photocatalysts. The precursors S^A (4,4'-biphenyl-2,2'-diylbis(2-hydroxy-1H-benzotriazole-4-yl)) and S^B (4,4'-biphenyl-2,2'-diylbis(2-hydroxy-1H-benzotriazole-4-yl)) were prepared as described in the Supporting Information (Figures S1–S5). Another monomer 2,4,6-trihydroxybenzene-1,3,5-tricarbaldehyde (TP) was synthesized according to the reported procedure (Figures S6 and S7).39

INPUTS

Select monomers, linkage type, and reaction conditions.

3D PREVIEW

Inspect monomer and candidate dimer structures.

ANALYSIS CHAIN

Follow staged checks from parsing to conclusion.

DEMO EXAMPLE

Video example: Tp + pPD/Pa, imine linkage, Schiff-base conditions, 3D preview, and an eight-step evidence chain analysis.

COF-System AI4S reviewer demo PDF

4

The monomer library supplies curated input records for design analysis and lets users inspect structures before using them in a workflow.



INSPECT

PREVIEW

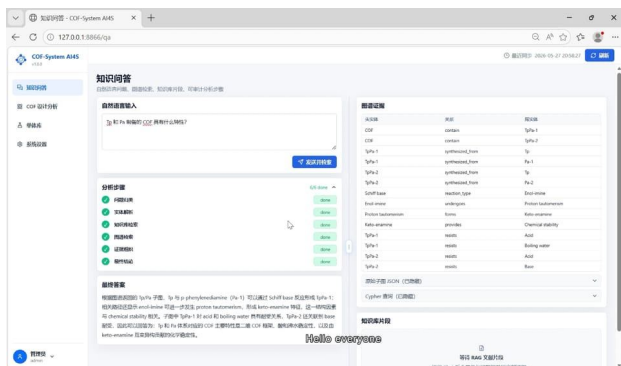
ROLE IN THE WORKFLOW

COF-System AI4S reviewer demo PDF

Video Walkthrough in Four Frames

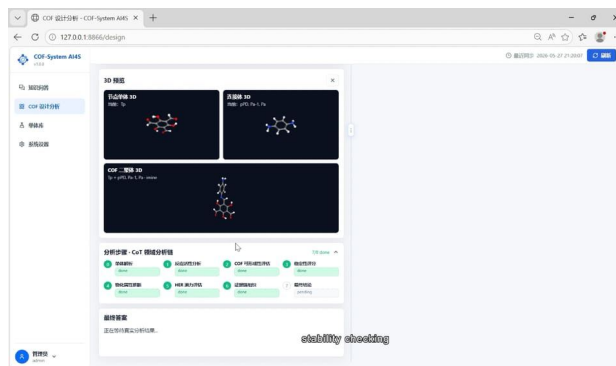
These frames are extracted from the 516-second demonstration video and summarize the reviewer's path through the interface.

01. Ask and retrieve



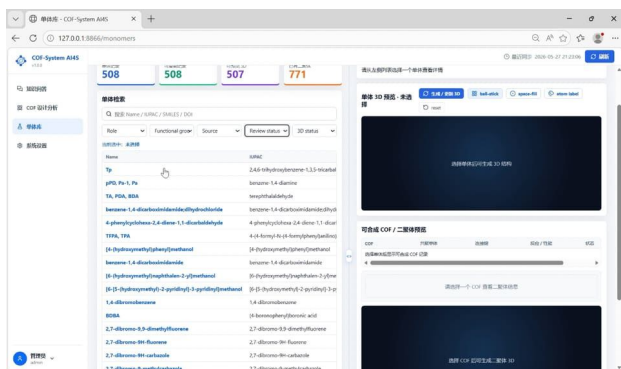
The QA page accepts a domain question and runs the retrieval workflow before displaying a final answer.

02. Preview chemistry



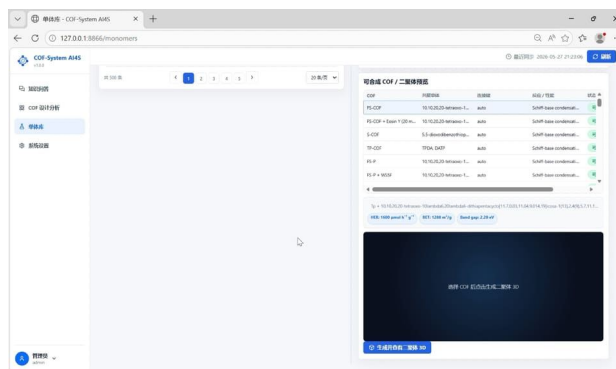
The design page previews selected monomers and a possible dimer before the analysis chain completes.

03. Browse monomers



The library exposes searchable records, functional groups, source fields, and available 3D status.

04. Inspect dimer 3D



Related COF/dimer candidates can be selected and generated for visual inspection.

REVIEWER QUICK GUIDE

The screenshots verify three claims: answers stay paired with graph and literature evidence; design feasibility is presented as a staged analysis chain; and monomer records are connected to 3D molecular previews and related COF/dimer candidates. The HER module should be read as an evidence-aware demo assessment.